

Accuracy is not enough

One number cannot tell you whether a model is useful. Two kinds of mistake, with very different costs, hide inside every accuracy figure. Today you learn to open them up — and meet a model that is right 98% of the time and helps nobody at all.

Unit 4 · Real data

80 minutes

Worksheet in pairs, not a lab

Bring a pen

The 98% model

Confusion matrix

The trade-off

Worksheet

Before Friday

PART 1 · 10 MINUTES

A 98%-accurate model that is useless

1

A screening programme tests 1,000 people for a condition that 20 of them have. A model answers *healthy* for every single person. It is right about 980 people: **98% accuracy**. It finds **none** of the 20 people the programme exists to find.

CLASS IMBALANCE

When one class is rare, a model can achieve high accuracy by ignoring it completely. **Always compare to the baseline of predicting the majority class** — here, 98% — and never report accuracy alone on imbalanced data.

The confusion matrix

Four numbers that accuracy squashes into one.

2

A district health post predicts which of 200 booked patients will miss their appointment, so a nurse can phone them the day before. Treat *misses* as the positive class:

EVERY CELL HAS A MEANING IN THE CLINIC'S OWN WORDS

	predicted: attends	predicted: misses
actually attended	169 true negatives attended, not flagged	1 false positive attended, but phoned
actually missed	26 false negatives missed, never phoned	4 true positives missed, and phoned

Measure	Question it answers	Here
Accuracy	what share of all predictions were right?	$(169 + 4) / 200 = \mathbf{0.865}$
Baseline	what does <i>everyone attends</i> score?	$170 / 200 = \mathbf{0.850}$
Precision	of the patients flagged, how many really missed?	$4 / 5 = \mathbf{0.800}$
Recall	of the patients who missed, how many were flagged?	$4 / 30 = \mathbf{0.133}$
F1	one number balancing the two	$2 \cdot \text{P-R} / (\text{P} + \text{R}) = \mathbf{0.229}$

Read it as the clinic would. When the model flags someone it is usually right (precision 0.8). But it flags almost nobody: 26 of the 30 patients who missed were never phoned. An accuracy of 86.5% sounded like success; a recall of 13% is the truth about what the model does for the clinic.

The threshold, and the trade-off

Nothing forces the cut-off to be 0.5.

3

The model outputs a probability. Calling someone a likely no-show when $p \geq 0.5$ is a convention, not a law. Move the cut-off and the two error types trade places:

Threshold	Patients flagged	Precision	Recall
0.1	117	0.222	0.867
0.2	46	0.391	0.600
0.3	18	0.444	0.267
0.4	9	0.556	0.167
0.5	5	0.800	0.133

CHOOSE THE THRESHOLD FROM WHAT MISTAKES COST

A phone call costs about 200 FCFA and five minutes of a nurse's time. A missed appointment wastes a consultation slot another patient needed. If a missed slot costs far more than a call, a low threshold — many calls, most no-shows caught — is right even though precision falls. **The costs belong to the clinic, not to the model.**

ROC and AUC

Sweep the threshold from 1 down to 0 and plot, at each setting, the **true-positive rate** (recall) against the **false-positive rate** (share of attending patients wrongly flagged). That curve is the ROC curve. The **area under it, AUC**, is the probability that the model ranks a randomly chosen no-show above a randomly chosen attender. 0.5 is a coin toss; 1.0 is perfect ranking. This clinic model scores **0.768**: it ranks reasonably well, even though the default threshold used that ranking badly.



Which number to report

AUC judges the ranking, independent of any threshold — good for comparing models. Precision and recall at a chosen threshold judge the decision you will actually deploy — what the clinic lives with. Report both, and always beside the baseline.

WORKSHEET · 30 MINUTES · IN PAIRS

Accuracy is not enough

4

The worksheet is `CSC4315_S21_Worksheet_Accuracy_Is_Not_Enough`, done on paper with a partner: read the matrix in the clinic's words, compute the four measures by hand, take apart the 98% model, and choose a threshold with a justification. The notebook holds check cells for afterwards.

Before Friday

1. **Without notes:** precision, recall, and one situation where each matters more.
2. **Look back at Lab 11's loss curves.** Friday asks what it means when training loss keeps falling and the other curve does not.



Session 22 — Friday 20 November, 11:30

Overfitting and what to do about it — training loss down, validation loss up, seen rather than described.